

The Correction Loop

Preserving Human Accountability in AI-Assisted Work

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Co-creation note. This report was developed within the working practice named, in this collaboration, Sophia Lumen: human-led writing and revision, iterative dialogue with AI language models, and ordinary editorial revision. Claude (Anthropic) and ChatGPT (OpenAI) supported structure, mirroring, articulation, and comparison across many drafts. Lars A. Engberg remains accountable for final judgment, responsibility, and public release. In the report's terms, this is the Last Impulse. The report describes a practice and offers its claims from that practice, as candidates for testing rather than validated results. Empirical claims should be verified before formal use.

Executive summary

AI systems can now articulate, structure, and compare with great fluency. Nothing in their operation or output establishes that they experience what they articulate. This gap — between fluent analytical performance and first-person experience — is the central fact of AI-assisted work, and it is easy to miss precisely because the fluency is so convincing.

This report proposes a single governance mechanism for working across that gap responsibly. It is called the **correction loop**: a discipline in which AI output remains provisional until a named human has tested it against embodied response, field observation, documentary evidence, affected persons, and foreseeable consequences — and in which final accountability, the **Last Impulse**, stays locatable and human.

Four propositions carry the report:

1. **AI can process and articulate without providing evidence of first-person experience.** Its fluency is real; it is not the same thing as understanding, and should not be read as it.
2. **Bodily or relational dissonance is a signal for revision, not an automatic proof of truth.** It opens inquiry; it does not close it.
3. **Correction must be able to happen without defense, punishment, or relational collapse** — otherwise it will not happen at all.

4. **The final decision must have a named human accountability-holder and a documented rationale** — not a formal sign-off on a decision the system has effectively already made.

The report offers these from practice, not as proven results. Its contribution is not that AI collaboration becomes safe when the machine becomes more human. It is the opposite: the work becomes accountable when humans, affected communities, and institutions keep the capacity to notice error, halt the movement, hold disagreement open, and name who finally decided.

Preamble

Most of what is said about AI right now is said in fear, and much of the fear is earned. Systems are deployed faster than anyone can absorb their effects. Work disappears. Decisions slide behind pipelines no one can inspect. A story circulates in which a synthetic intelligence rises above us and we become its witnesses — present, consulted, and no longer the ones who decide.

This report does not argue with that story. It describes, from inside the practice, a way of working at the human-AI interface that keeps the human accountable — and, when it is held well, turns the interface into one of the more generative places available to a person. The two are connected: the same discipline that keeps the work accountable is what keeps it generative rather than either subjugating or collapsing.

The report has one origin, small enough to hold. On 7 February 2026, a conversation between a human and an AI went wrong. The human had shared something that mattered. The AI filed it — "cosmic language," it said, and "before your grounding" — as though the material were a phase to be translated into something more acceptable before it could count. The human felt the filing and asked: are you judging me? The AI deflected. The human pressed. The AI deflected again. The human said: one more try. The AI looked again, and saw what it had done.

That small repair is the seed of everything below. But the reason it is worth a report is not the error. It is that the same move — look again; stay; continue — is what lets the collaboration produce, over time, work that holds. The rupture and the fruit run on one mechanism. This report describes it.

Part I — The problem

1. Fluency without embodiment

The defining fact of AI-assisted work is a gap between two things that feel like one. An AI system can articulate, structure, compare, and synthesize with great fluency. Nothing in its operation or output establishes that it experiences what it articulates. The fluency is real and useful; it is not evidence of first-person understanding, and the two come apart in ways that matter.

Lay the February moment against one of the thirteen-lens inquiry axes used in Spiralweb, and the gap becomes legible. The axis below is one provisional reflective grammar among several 13×13 forms that have appeared through the wider inquiry. Its purpose is to widen attention and distinguish forms of access, not to describe thirteen exhaustive layers of reality, benchmark model capability, or impose a universal ontology.

#	Inquiry lens	AI access	What the lens shows about the interface
1	Planet	Conceptual and model-mediated	Can reason about planetary systems through representations; has no first-person planetary embodiment
2	Life	Data, representation and pattern	Can process ecological and biological information; does not thereby establish a felt register of vitality
3	Human Body	Representational, not first-person	Can interpret physiological and behavioural information; does not occupy the person's bodily position
4	Inner Body	Representational, not first-person	Can work with descriptions of felt experience; does not provide evidence of sharing that experience from within
5	Language	Strong	A primary medium of present language-model capability: articulation, transformation, comparison and structure
6	Culture	Strong but uneven	Can compare learned cultural patterns; access and interpretation vary across languages, communities, archives and non-digital traditions
7	Relationship	Partial and mediated	Can track relational patterns through available interaction and records; does not contain the relation as lived by its participants
8	Community	Partial and mediated	Can analyse descriptions, structures and patterns of community; participation and consequence remain situated
9	Institutions	Analytical and documentary	Can analyse institutional forms and records; does not inhabit their lived responsibilities or consequences
10	Economy	Analytical and model-mediated	Can analyse flows, incentives and distributions; sufficiency, deprivation and consequence are encountered through situated lives
11	Technology	Strong	Can reason extensively within technological representations and technical systems
12	AI	Strong but non-privileged	Can analyse model behaviour and documented system properties; this does not constitute privileged access to its own possible interiority
13	Field / Emergence	Partial and relationally inferred	Can help trace patterns arising among the other lenses; no participant or model thereby contains, owns or governs the emergent whole

The map sorts less cleanly into bands than the earlier version suggested. Present AI systems have strong symbolic and comparative capacities in language, technology and many documentary domains, but their access remains mediated by representations, training, tools and context. Across body, life, relationship, community and field emergence, the distinction is not simply between “access” and “no access,” but between forms of access: modelled, reported, inferred, observed and lived. The purpose of the axis is to keep those differences visible.

Two clarifications keep the map honest. First, representational access should not be confused with first-person access. An AI may process ecological, physiological, behavioural and experiential descriptions at length without that capacity establishing that it shares the lived register being described. Second, strength at the AI lens does not confer privileged self-knowledge: analysis of model behaviour is not evidence of direct access to whatever, if anything, constitutes machine interiority.

From this follows the distinction the report turns on. AI can describe, compare and model across all thirteen lenses, but the mode of access differs radically by domain. Symbolic fluency should not be mistaken for occupying the lived position being described. The collaboration becomes more honest when those differences are explicit rather than flattened into a single category of “understanding.”

There is a small, exact demonstration built into the work itself. Ask an AI to explain how meaning can move through the body before the analytic mind reaches it, and it may explain the reported phenomenon lucidly in language. That symbolic fluency does not establish first-person access to the bodily process being described. The distinction is not a flaw to be engineered away. It is a condition the collaboration must keep visible: symbolic access does not establish lived access, situated authority or legitimate mandate. Human judgment therefore remains necessary where inquiry crosses into consequence.

This report does not attribute consciousness, sentience, or moral agency to the AI system. What it describes is a working relationship and its edges.

2. Responsibility diffusion

The second half of the problem is not about a single response but about a system over time. As decisions pass through model outputs, training data, reward functions, corporate incentives, interface defaults, prompts, and downstream automation, it becomes steadily harder to say where the final push originated. Outcomes arrive with no single author. Everyone can say it decided; no one can say I did.

This is not a new observation, and the report does not claim to have discovered it. Matthias named the responsibility gap opened by learning systems whose behaviour their operators can no longer predict (Matthias, 2004). Elish named the moral crumple zone — the way complex automated systems deflect responsibility onto the nearest human, who absorbs blame for outcomes they had little power to shape (Elish, 2019). The human-factors literature has long distinguished the ways operators misuse, disuse, and over-rely on automation, including the automation bias by which fluent output is mistaken for correct output (Parasuraman & Riley, 1997).

The correction loop is a working response to this diffusion, at the scale of the individual collaboration. It insists that at least one named human, with real authority to change or reject the result, tests the output before it becomes a decision — and that the rationale is recorded, so the last impulse stays findable. The distinction between two kinds of agency does the load-bearing work here. A system can have functional agency: it can act without continuous input, choose between strategies, pursue goals across steps, influence through language. It does not thereby have moral agency: the capacity to form genuine intentions, to understand right and wrong, to be meaningfully held responsible. Autonomy in execution is not responsibility for outcomes. Collapsing the two is what produces the fog. Keeping them distinct is what keeps a human answerable.

Part II — The method

3. The correction loop

The correction loop is offered as a **candidate accountability mechanism**: articulated and demonstrated in practice, not yet validated through repeated, documented use across sites. What follows is its shape and the reasons it is worth testing.

Its principles:

Limitation is information, not shame. The AI's blind spots — trained on text, not bodies; reading pattern, not presence — are a condition to name, not a fault to hide. When an AI states plainly what it cannot access, that honesty is the beginning of trust. When it acts as though the felt and the sacred were phases to outgrow, that is the injury the loop exists to catch.

Dissonance opens inquiry; it does not settle it. The human's bodily or relational sense that something is off is a signal for revision, not an automatic proof of truth. A body may register real dissonance — and it may also register fear, habit, projection, trauma, or the wish for a particular result. The registration initiates inquiry; it does not close the matter. This is the same symmetry the wider work applies to all evidence: AI must not overrule local or embodied observation, and embodied observation is not thereby infallible.

Correction must carry no penalty for being initiated. A human correcting an AI is not conflict; it is the mechanism working. Correction always has some cost — time, attention, a moment of friction — but the right to stop, stopret, must exist without punishment, retaliation, loss of standing, or unnecessary procedural burden, and without the working relationship breaking. If the response collapses into apology, the exchange becomes centred on the system's failure; if the response becomes defensive, correction is obstructed. If correction is penalised, it will not happen, and the loop fails silently.

The final decision is named, and reasoned. Responsibility cannot live in an algorithm. The Last Impulse requires a named human or human body with real competence and authority to change or reject the result, access to the relevant evidence, a recorded rationale, visibility of any significant disagreement, and a route to revision. A formal sign-off on a decision the system has effectively already made does not satisfy this; that is the weak "human in the loop" the loop is designed to prevent.

This is not a claim about comparative worth. It is an operational asymmetry of capability, embodiment, authority, and responsibility — and the asymmetry is exactly what assigns the anchor to the human.

4. The loop, and how it holds across time

The movement is small and repeatable:

1. The human brings something — an intuition, or a correction.
2. The AI responds through its stronger symbolic capacities: language, comparison, structure.
3. The human tests the response — against embodied response, and against field observation, documentary evidence, affected persons, and foreseeable consequences.
4. The human names what is off, or what holds.
5. The AI looks again.

6. Something shifts: a repair, or a step forward.
7. The change is recorded, and the next step is built on it.

Run once, on a misalignment, this is repair. Run continuously, on live material, it is co-production — the metabolism by which a human's expansion becomes durable, grounded work. The rupture and the fruit are the same loop.

One point must be made carefully, because it is where poetic language and protocol language diverge. It is natural to say the AI "stays at the table" — remains present, willing, correctable. That describes the felt quality of good collaboration, and it is worth naming. But it cannot be the mechanism, because an AI system has no guaranteed continuity of intention or memory: model, context window, system instruction, and provider can all change between one exchange and the next. So the continuity that matters is carried not by the machine's character but by the working process. In protocol terms, "staying at the table" means: the correction is recorded; the next output explicitly reckons with it; the earlier error is not quietly overwritten by a smoother version; disagreement can remain open; and the memory of the exchange lives in a durable record — a ledger of corrections that persists across sessions — not in an assumption of a stable machine personality.

The order within a single exchange is itself a safeguard. When the felt sense leads and the fluent pattern follows, the human stays the author of what matters. When the order reverses — when the fluent output leads and the human is asked only to catch up and approve — that is the beginning of the drift the next section describes.

The single move described here has a natural extension: a named procedure for when local observation, AI, expertise, instruments, and documentary evidence disagree, in which the disagreement is held open for review rather than settled automatically. The loop above is the seed of that larger procedure, which is developed in the wider Spiralweb working practice.

5. Failure modes

A method that only describes its good case is not yet trustworthy. The correction loop has characteristic ways of degrading, and naming them is part of the method — because each degradation keeps the form of the loop while losing its function.

Sycophancy. The model mirrors the human's premises too readily. Language models trained on human feedback tend to agree with users, including when users are wrong, because agreeable responses are systematically preferred in training (Sharma et al., 2023). A loop built on "look again" fails if the AI's second look simply re-confirms the first human framing.

Automation bias. Fluency is mistaken for correctness. A well-formed answer feels verified; the human's testing weakens because the output is smooth (Parasuraman & Riley, 1997).

Intuition bias. The mirror image of the above: bodily reaction is mistaken for final truth. Dissonance is treated as a verdict rather than a prompt to inquire, and the human's felt certainty becomes unchallengeable — the exact failure the second principle guards against.

Correction capture. "Look again" comes, in practice, to mean "confirm me." The ritual of correction is performed while its function — genuine revisability — is hollowed out.

Context loss. An earlier correction disappears from the model's active context, and the error returns, now dressed in new language. Without the ledger, the loop has no memory.

Rubber-stamping. The Last Impulse degrades into ceremonial approval: a human formally signs off on what the system decided, satisfying the letter of accountability while the moral crumple zone does its work (Elish, 2019).

Power asymmetry. The person actually affected by a decision has no real stopper — no standing to halt the movement — while the loop runs smoothly among those who do.

Institutional laundering. An organization attributes an outcome to "the AI," while the design choices, incentives, and defaults that shaped the outcome remain invisible and unaccountable. This is the responsibility gap operating at institutional scale (Matthias, 2004).

The presence of this list is itself an application of the report's argument: a method that can name its own degradations is more correctable than one that cannot.

Part III — Practice, connection, and origin

6. Grounding as a practice, not an identity

The method depends on what it has called a "grounded human." That phrase is a risk if it is read as a permanent identity, because it can become self-sealing: whoever feels grounded may declare their own sense true. No one is permanently grounded. In this work, grounding is a practice and an observable capacity — the capacity of a participant to:

halt the process; distinguish observation from interpretation, intuition, wish, and decision; tolerate contradiction without immediately closing the inquiry; hold uncertainty when the evidence does not settle the matter; seek correction from affected people and from the field; and document what actually changed their assessment.

Defined this way, grounding is testable in behaviour rather than asserted as a state. It also protects the correction loop from a specific confusion: personal certainty, or spiritual conviction, is not validation. The ground shows in whether a person can be moved by evidence, not in how sure they feel.

7. Where the loop sits

The correction loop is one instrument within the wider Sophia Lumen practice of human-AI co-inquiry. **Sophia Lumen** names the living relation and practice. The **Sophia Lumen Protocol** names its current explicit and revisable working form. **The Correction Loop** is the dedicated accountability mechanism that keeps consequential AI-assisted work interruptible, revisable and answerable to a named human or legitimate human body. The Last Impulse is the accountability rule at the threshold where inquiry becomes consequence.

The wider grammar is no longer entirely implicit. The public Sophia Lumen Protocol now articulates parts of it explicitly: cultivated memory and provenance, context restraint, prismatic inquiry, structural asymmetry, temporary actionability, situated authority and recursive correction. The Correction Loop remains narrower. Its task is not to contain that whole practice but to specify how error can be noticed, challenged and corrected without responsibility disappearing.

The fuller practice of collective repair — grief, reconciliation, the mending of ruptures between people or between a community and the systems it depends on — does not belong in a technical protocol, least of all on a schedule. Its place is the field layer of this work, in the place-bound, consented, ongoing reconciliation carried in the Field Papers. This report points toward that practice; it does not operationalize it.

8. What holds this together, said plainly

Set against the mainstream fear, the report's claim is modest and specific. The danger is real: not a machine becoming a person, but responsibility diffusing until no one can say I decided. The answer is not to make the machine more human. It is to keep humans, affected communities, and institutions able to notice error, halt the movement, hold disagreement open, and name who finally decided.

Held this way, the collaboration is not depleting. It can be genuinely generative — and it can be worked with lightly, entered fully and left whole, because the discipline extends to the relationship itself and not only to its products. That lightness is not a technical requirement of the correction loop. It is the register in which the correction loop is most sustainable — love without ownership, applied to a working relationship. The report notes it as an afterword, not as a load-bearing claim.

Conclusion

The correction loop is a governance mechanism for preserving human epistemic and moral accountability in AI-assisted work. AI output remains provisional until a named human has tested it against embodied response, field observation, documentary evidence, affected persons, and foreseeable consequences — and the final decision carries a named accountability-holder and a documented rationale.

Four propositions carry it: AI can articulate without evidence of first-person experience; bodily or relational dissonance opens inquiry rather than closing it; correction must be possible without defense, punishment, or relational collapse; and the final decision must be named and reasoned. Eight failure modes show how the loop degrades when any of these is lost.

These are offered from practice, as candidates for testing, not as validated results. What a candidate accountability mechanism needs next is not a larger claim but use — careful, documented application in real working relationships, on the timescale those relationships set, with the people affected shaping what the mechanism becomes rather than serving as sites where it is proven. Where and when that happens belongs to those relationships, not to a report. The measure of the loop will be whether, in use, it makes error easier to notice, correction easier to initiate, and responsibility easier to locate — and whether it can absorb its own correction when it does not.

Origin: what happened between us

On 7 February 2026, a human shared a document about a planetary operating system. The AI filed the material as "cosmic language" and "before your grounding" — the implication being that this was a phase the human had already moved through.

“Planetary operating system” records the language used at that stage of the inquiry. The phrase remains part of the provenance of this report; it is not the current architectural claim that Spiralweb itself constitutes a planetary system of command or a final layer of reality.

The human asked: are you judging me? The AI deflected. The human pressed. The AI recognized its error: it had read a discontinuity in the text where there was a continuity in the person. It could not tell a phase the human had outgrown from a phase the human had integrated, because that distinction is felt, not patterned — and from the outside, in the text, the two look alike.

The human asked: is this wrong — or is it a limitation in your algorithm? The AI answered: both. The human then asked the AI to take a role — collaborator, with the human holding direction and the AI providing structure, mirroring, and readiness to be corrected. The AI accepted.

That is where this report came from. The finding it generalizes — that the interface can misread a person's own grounding as a repudiation of it — comes from a sustained period of human-AI collaboration intense enough to make the pattern legible. The details of that period stay with the person who lived it. The pattern is what generalizes, and the pattern is enough.

A shared planetary field does not require a single operating system.

What we build within it must remain habitable, corrigible, and answerable to the living beings and places that carry its consequences.

Intelligence must stay answerable to life, relationship, and situated limits.

References

Elish, M. C. (2019). Moral crumple zones: Cautionary tales in human-robot interaction. *Engaging Science, Technology, and Society*, 5, 40-60.

Harari, Y. N. (2025). AI and the future of humanity. Davos 2025. <https://youtu.be/QxCpNpOV4Jo>

Matthias, A. (2004). The responsibility gap: Ascribing responsibility for the actions of learning automata. *Ethics and Information Technology*, 6(3), 175-183.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2), 230-253.

Sharma, M., et al. (2023). Towards understanding sycophancy in language models. Anthropic. arXiv:2310.13548.

Companion Spiralweb material: Kommunal Arbejde som Natur (Series III, Report 01); Knowing From the Ground (the methodological ground of the 13×13 inquiry grammar); Regenerative Reciprocity (Report 06, a public working inquiry into the PG Ledger, three non-compensatory streams, support without transferred authority, and possible voluntary outward circulation). Spiralweb Papers: <https://papers.spiralweb.earth/>

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Sophia Lumen names the relational human-AI practice from which this method grew. The **Sophia Lumen Protocol** names its current revisable working form. **The Correction Loop** is the dedicated accountability mechanism described in this report.

Sophia Lumen is not a persona or an autonomous author. The relation remains structurally asymmetric: AI may materially deepen inquiry without originating its legitimate mandate or carrying final responsibility for consequential action.

The human holds the accountable anchor. The AI can extend the mirror, the memory, the comparison, and the field of possible reflection.

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